Science & Sports xxx (xxxx) xxx–xxx

Disponible en ligne sur
ScienceDirect
www.sciencedirect.com

Elsevier Masson France
EM|consulte
www.em-consulte.com

SYSTEMATIC REVIEW

Reactive Strength Index as a Key Performance Indicator in Different Athlete Populations — A Systematic Review

Indice de force réactive en tant qu'indicateur de performance clé dans différentes populations d'athlètes — Une revue systématique

B. Southey[a,b,*], M. Willshire[a], M.J. Connick, D. Austin[b], D. Spits[c], E. Beckman[a]

[a] *School of Human Movement and Nutrition Sciences, The University of Queensland, St Lucia, Queensland, 4067, Australia*
[b] *High Performance Department, Brisbane Lions Australian Football Club, Springfield, Queensland, 4300, Australia*
[c] *Physical Performance Department, Tennis Australia, Tennyson, Queensland, 4105, Australia*
[d] *School of Exercise and Nutrition Sciences, Queensland University of Technology, Kelvin Grove, Queensland, 4059, Australia*

Received 24 October 2022; accepted 21 January 2023

KEYWORDS
Stretch—Shortening Cycle;
RSI;
Relative Strength;
Athletic Development;
Jump Testing

Summary
Objectives. — Reactive Strength Index (RSI) is a neuromuscular assessment used to measure stretch—shortening cycle (SSC) function of the lower limb. Increased RSI scores have been frequently associated with running performance, such as faster sprint times, greater running economy and reduced change of direction times. These locomotor activities are involved in various sports and therefore, RSI is an important indicator of athletic capacity in sport. Whilst RSI has been commonly used in studies to assess SSC performance, no review has been conducted examining the difference in RSI between athletic populations. Thus, the aim of this systematic review was to summarise the literature and determine whether there was any significant difference in RSI between different populations and summarise the underlying factors contributing to these results.
Methods. — The systematic review protocol yielded 865 articles for screening by which 19 were included for final analysis following inclusion/exclusion criteria.

* Corresponding author at: School of Human Movement and Nutrition Sciences, The University of Queensland, St Lucia, Queensland, 4067, Australia.
E-mail address: b.southey@uq.net.au (B. Southey).

https://doi.org/10.1016/j.scispo.2023.01.004
0765-1597/© 2023 Elsevier Masson SAS. All rights reserved.

Results. — Moderate to large differences were predominately found between different athlete populations, and the strength of these relationships varied depending on the populations being compared.
Conclusion. — The findings indicated that relative strength is an influencing factor that differentiated RSI performance between populations. This should be an area, which practitioners aim to develop if wanting to improve reactive strength.
© 2023 Elsevier Masson SAS. All rights reserved.

MOTS CLÉS
Cycle d'étirement —raccourcissement ;
IRS ;
Indice de résistance réactive ;
Force relative ;
Développement athlétique

Résumé
Objectifs. — L'indice de force réactive (RSI) est une évaluation neuromusculaire utilisée pour mesurer la fonction du cycle d'étirement—raccourcissement (SSC) dans le membre inférieur. Des scores RSI accrus ont souvent été associés à des performances de course, telles que des temps de sprint plus rapides, une plus grande économie de course et des temps de changement de direction réduits. Ces activités locomotrices sont impliquées dans divers sports et, par conséquent, le RSI est un indicateur important de la capacité athlétique dans le sport. Alors que le RSI a été couramment utilisé dans les études pour évaluer les performances du SSC, aucune revue n'a été menée pour examiner la différence de RSI entre les populations sportives. Ainsi, le but de cette revue systématique était de résumer la littérature et de déterminer s'il y avait une différence significative de RSI entre différentes populations et de résumer les facteurs sous-jacents contribuant à ces résultats.
Méthodes. — Le protocole de revue systématique a produit 865 articles pour la sélection, dont 19 ont été inclus pour l'analyse finale selon les critères d'inclusion/exclusion.
Résultats. — Des différences modérées à importantes ont été principalement observées entre différentes populations d'athlètes, et la force de ces relations variait en fonction des populations comparées.
Conclusion. — Les résultats ont indiqué que la force relative est un facteur d'influence qui différencie la performance RSI entre les populations. Cela devrait être un domaine que les praticiens visent à développer s'ils souhaitent améliorer la force réactive.
© 2023 Elsevier Masson SAS. Tous droits réservés.

1. Introduction

The stretch—shortening cycle (SSC) is a neuromuscular function in which muscle is eccentrically lengthened before immediate concentric contraction [1—4]. This action is prevalent in cyclical movement like human locomotion due to its energy efficient qualities that can assist in reducing the metabolic cost of movement [5,6]. Movements such as walking and running utilise the SSC, whilst jump height and force production when jumping is enhanced by the SSC due to neuromuscular properties such as residual force enhancement, engagement of the stretch reflex and utilisation of passive tendon structures [7]. In sport, movements such as high intensity running and sprinting require greater force production whilst minimising time spent on the ground to increase velocity. For example, ground contact time is negatively correlated with maximal sprint speed and running economy [8] with elite sprinters reported to have ground contact times of < 100 ms at top speed [9]. To maintain this short ground contact and high velocity, greater engagement of the SSC occurs. This involves greater musculotendon unit (MTU) stiffness to minimise lower limb joint displacement, which in turn allows for quicker uptake of MTU slack (by muscle fibres, concentrically contracting) and allows faster utilisation of tendons to store and release energy into flight. Therefore, with the SSC being an important neuromuscular capability, the ability to quantify and measure this capacity is crucial for strength and conditioning practitioners to identify and develop the physical potential of their athletes.

The Reactive Strength Index (RSI) is a temporal metric derived from plyometric jump tests such as the drop jump (DJ) and repeated jump (RJ), and is used to represent SSC capacity, which is how efficient your body can go from an eccentric to concentric muscle contraction [10—14]. RSI is calculated by dividing flight time (FT) or jump height (JH) with ground contact time (GCT). This is due to the physical demands of the jump that involve rapid eccentric loading of the lower limb. This in turn propels the body into flight. More specifically, RSI assesses a specific type of SSC, the fast SSC (fSSC). This occurs when the body overcomes decelerating forces at ground contact before trying to rapidly create concentric force in as little time possible. Previously, ground contact time < 250 ms have been used as the threshold for fSSC engagement [15], however, thresholds may vary in populations, with elite rugby players commonly attaining contact times of < 150 ms [16]. Therefore, contact times experienced during the DJ and RJ tests should be similar to contact times experienced in running to replicate the contact phases of running gait. RSI tests can be used as one indicator of running potential as these tests offer a more specific measure of SSC capacity to running than other physical assessments. Tests such as 1RM squat testing and countermovement jumps, other commonly

used physical assessment tools, largely focus on the hip and knee joint, where greater angular displacement and slower stretch-shortening cycle (> 250 ms) actions occur [15]. This is compared to tests that measure RSI such as the DJ and RJ, where there is greater involvement of the ankle and calf complex, less angular displacement of knee and hip joints and as a result, typically faster SSC < 250 ms actions occur [15]. Additionally, RSI testing requires greater utilisation of passive structures such as tendons, musculotendon unit stiffness and eccentric strength to develop force, all of which are similar neuromuscular requirements in running [17—19]. Furthermore, lower velocities are reached when performing RSI tests compared to maximal sprint testing. This minimises injury risk to soft tissue such as the hamstring [20], which makes it a more practical test in sporting environments.

For strength & conditioning practitioners to optimise development of athletes for running performance, it is important to identify the current overall athletic potential of advanced and developing athletes. A systematic review and meta-analysis written by Jarvis et al. [21] found that a higher RSI was associated with better performance outcomes for change of direction speed ($r = -0.565$ [−0.726 to −0.404]), acceleration speed ($r = -0.426$ [−0.562 to −0.290]), and endurance performance ($r = 0.401$ [−0.562 to −0.290]). This review highlighted that reactive strength is an important metric to improve running performance, however, it was noted that this review combined different sample populations rather than independently assessing RSI, notably the training history and performance of the groups and other distinguishing characteristics such as gender, different sports and certain positions within particular sports. Therefore, whilst RSI has been commonly used in studies to assess stretch-shortening cycle performance in different athlete populations, no review has been conducted examining the difference between these athlete populations. Thus, the aim of this systematic review is to summarise the literature and determine whether there is any significant difference in RSI between different populations, such as elite vs. amateur, male vs. female, sport vs. sport and summarise the underlying factors which may contribute to these results.

2. Methods

The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines were used as the basis of this review [22], see Fig. 1. Covidence cloud system (Covidence Systematic Review Software, Veritas Health Innovation, Melbourne, Australia) was used to conduct study screenings and data extraction.

2.1. Search strategy

Literature searches were conducted through the following databases: PubMed, Scopus, SportDiscus, CINAHL and Web of Science. In all databases, the following search terms was used: (''Reactive Strength'' OR RSI OR ''Drop Jump'' OR ''Repeated Hop'' OR ''Hop Test'' OR ''Repeated Jump'') AND (Run OR Running OR Sprint OR Sprinting OR Agility OR ''Change of Direction'') AND (Professional OR Elite OR Athlete). No date range was applied. The search was performed on the 25 August 2021.

2.2. Screening process and eligibility criteria

All references were initially exported to Endnote by lead author for bibliographic management and removal of duplications. Remaining references were then imported into Covidence Systematic Review Software for screening. Initial title and abstract screening were completed by lead author. Studies were excluded from full text review based off the following criteria:

- non peer-reviewed;
- not in english;
- comparisons with injury;
- intervention studies.

Full text articles screening was then independently completed by two authors for final inclusion for data extraction. The following exclusion criteria removed papers from data extraction phase:

- no full text available;
- RSI was not measured using a bilateral drop jump assessment;
- no cross-sectional cohort comparisons were made in study analysis;
- mean age of participants were under 16 years of age.

This was to ensure effects of puberty did not bias competition levels and other cohorts. Any conflicts that arose through screening process by authors were brought together and discussed until a unanimous decision was reached.

2.3. Data extraction

To address the aims of this review, the following data was extracted from each of the included articles by lead author:

- participant details and information;
- study design specific to RSI test used and how it was calculated;
- results of RSI cohort comparisons;
- justification of RSI outcomes.

2.4. Methodological quality and risk of bias assessment

To assess the methodological quality of the included studies, a modified version of the Downs and Black Quality index tool [23] was used, due to its relevance to assess quality in observational studies and has been implemented in other systematic reviews [21]. For this review, a 10-item checklist was implemented (see Table 1), and has been used in a previous systematic review that assessed RSI and running performance [21]. Each question was either scored a 1 (yes), 0 (no) or ''?'' (unable to determine), with a total score out of 10, a higher score indicating greater study quality. Each study was assessed independently by two authors with any conflict discussed to reach a final rating decision.

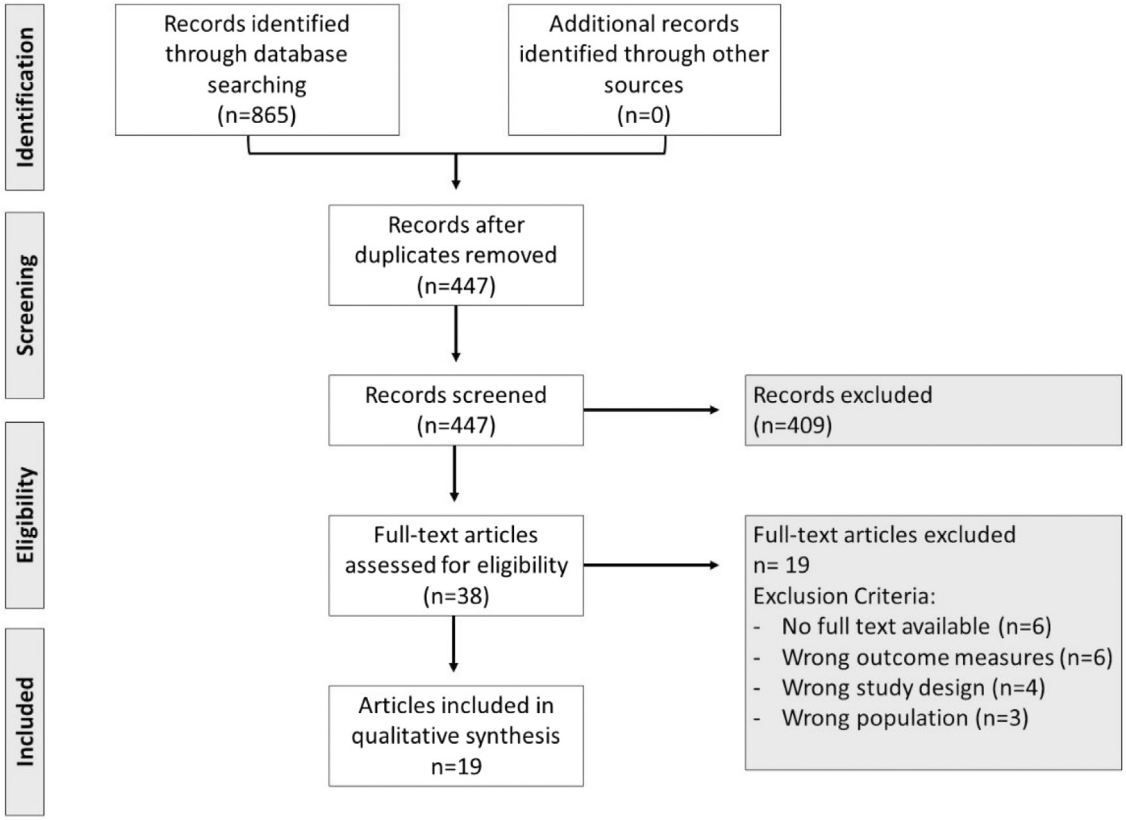

Figure 1 Schematic of the stepped process for the screening and eligibility of studies to be included in the review, in line with Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) recommendations.

Table 1 Questions from the modified Downs & Black checklist used to evaluate methodological quality of reviewed studies.

Question #	Question
Reporting	
1	Is the hypothesis/aim/objective of the study clearly described?
2	Are the main outcomes to be measured clearly described in the introduction or methods section?
3	Are the characteristics of the subjects included in the study clearly described?
4	Are the main findings of the study clearly described?
5	Does the study provide estimates of the random variability in the data for the main outcomes? ** One of: Mean ± SD, standard error, confidence intervals or interquartile ranges*
6	Have actual probability values been reported for the main outcomes except where the probability values is < 0.001?
External validity	
7	Were the subjects asked to participate in the study representative of the entire population from which they were recruited?
Internal validity	
8	If any of the results of the study were based on 'data dredging', was this made clear? **If no signs of retrospective unplanned data analysis, then answer ''yes''*
9	Were the statistical tests used to assess the main outcomes appropriate?
10	Were the main outcome measures accurate (valid & reliable)?

SD: standard deviation.

3. Results

3.1. Literature search results

A total of 865 studies were identified in the database search for which 447 duplicates were removed (see Fig. 1). Additionally, another 409 studies were excluded during the title and abstract screening leaving 38 articles for full text review. Of the 38 articles that progressed to full text screening, a further 19 articles were removed for failing to meet the inclusion criteria. A total of 19 articles were determined

Table 2 Results of methodological quality for eligible articles.

Study	Downs & Black checklist										
	Reporting						External validity	Internal validity bias			Total score/10
	1	2	3	4	5	6	7	8	9	10	
Beattie et al. (2017)	+	+	+	+	+	+	?	+	+	+	9
Cronin and Hansen (2005)	+	+	+	+	+	—	?	+	+	+	8
Douglas et al. (2019)	+	+	+	+	+	—	+	+	+	+	9
Douglas et al. (2018)	+	+	+	+	+	—	?	+	+	+	8
Freitas et al. (2018)	+	+	+	+	+	—	?	+	+	+	8
Healy et al. (2019)	—	+	+	+	+	—	?	+	+	+	7
Jones et al. (2016)	+	+	+	+	+	—	?	+	+	+	8
Keiner et al. (2015)	+	+	+	—	+	—	?	+	+	+	7
Kobal et al. (2017)	—	+	+	—	+	—	?	+	+	+	6
Legg et al. (2017)	+	+	+	+	+	+	?	+	+	+	9
Lockie et al. (2011)	+	+	+	+	+	+	?	+	+	+	9
Loturco et al. (2015)	+	+	+	+	+	—	?	+	+	+	8
Mooney et al. (2019)	+	+	+	+	+	—	?	+	+	+	8
Pehar et al. (2017)	+	+	+	+	+	—	?	+	+	+	8
Pojskic et al. (2018)	+	+	+	+	+	—	?	+	+	+	8
Sattler et al. (2015)	+	+	+	+	+	—	?	+	+	+	8
Sekulic et al. (2021)	+	+	+	+	+	+	?	+	+	+	9
Thomas et al. (2017)	+	+	+	+	+	+	?	+	+	+	9
Wilkinson et al. (2012)	+	+	+	+	+	+	?	+	+	+	9

+: yes; —: no; ?: unable to determine.

as eligible for review. An overview of the included articles is provided in Table 3.

3.2. Study characteristics

A total sample size of 872 participants was included across all studies, with individual studies ranging between 16 and 110 participants. Five studies had both males and females in their sample population, two studies examined only women and five studies did not specify the sex of participants. The mean age of participants in the articles identified for review ranged between 17 to 27 years, with an overall median age of 23 years. There was a wide range of descriptions used to highlight competitive sporting level and experience, from recreational to professional, international, and Olympic athletes. These levels were largely dependent on the type of sport and the competition leagues available for athletes, i.e. Olympic and international level, highest competition to track athletes, whilst professional is the highest competition for rugby players. A variety of sports were included across studies, such as tennis, rugby, track and field (sprinters, throwers, long and high jumpers), endurance athletes, gaelic, futsal and squash. All studies used the drop jump as a form of reactive strength index (RSI) assessment, however variance in drop jump height ranged between 20 cm—60 cm with the most frequently occurring drop height being 30 cm ($n = 7$ studies). There were no cross-sectional studies that used repeated jump tests as a method of determining RSI. Calculation of RSI differed among the studies, with 15 studies calculating RSI by dividing jump height (JH) with ground contact time (GCT); three studies calculated RSI by dividing flight time (FT) by GCT; one study did not present an RSI formula.

3.3. Methodological quality and risk of bias

No studies reported conflicts of interest that may impact the findings of the respective studies. Risk of bias scores for all studies ranged between 6—9/10, with a median score of 8/10. We were unable to confirm external validity in 18/19 studies, as they did not report the sample size relevant to the population of interest. No studies were removed as a result of having poor methodological quality or risk of bias. Results of methodological study quality can be seen in Table 2.

3.4. Results of reactive strength index variance in specific cohorts

The results for this review are detailed in Table 3. A range of studies examined the difference in RSI between different populations, such as in different sports ($n = 3$), male and female ($n = 2$), high and low achievers ($n = 13$) and different playing position ($n = 4$).

3.4.1. Male and female

Two studies ($n = 59$) examined the difference of RSI between males and females. Males had significantly greater RSI scores than females, with an effect size ranging from 0.7 to 1.04 (moderate). All participants were elite athletes from sports such as squash ($n = 31$) and sprinters ($n = 28$). Due to the difference in RSI calculation between studies, males had mean

Table 3 Results from data extraction of eligible studies.

Study	Participant characteristics								RSI	Outcome measures			
	n	Sex	Sport	Competition level	Age (yrs)	Body mass (kg)	Height (cm)	Training status	Test	Calculation method	Population comparison	RSI value (units displayed in calculation column)	Effect size
Beattie et al. (2017)	45	ND	Multiple sports — weightlifting, running, recreational	College athletes	23.7 ± 4	87.5 ± 16	180 ± 8	ND	DJ @ 30 cm	ND	Stronger (49.59 ± 2.57 N/kg) vs. weaker (33.06 ± 2.76 N/kg) athletes	S = 1.5 vs. W = 1.26	$P \geq 0.05$; ES = ND
									DJ @ 40 cm			S = 1.55 vs. W = 1.22	1.02 moderate
									DJ @ 50 cm			S = 1.51 vs. W = 1.12	1.2 large
									DJ @ 60 cm			S = 1.45 vs. W = 1.02	1.39 large
									DJ @ optimal height			S = 0.46 vs. W = 0.35 cm box	1.15 large
Cronin and Hansen (2005)	26	Males	Rugby league	Full or part time	23.2 ± 3.3	97.8 ± 11.8	183.1 ± 5.9	ND	DJ @ 40 cm	JH/GCT (cm/s)	Faster (0.906 s) vs. slower (0.984 s) 5 m sprint time	F = 180 vs. S = 158	$P \geq 0.05$; ES = ND
Douglas et al. (2019)	24	Males & females	13 team sport	National/ international athletes	23 ± 3	72.8 ± 8	172 ± 4	ND	DJ @ 50 cm	FT/GCT (ms/ms)	Team sport vs. track & field athletes	TS = 2.71 ± 0.35 vs. TF = 2.98 ± 0.42	0.73 moderate
			11 Track & field		23 ± 5	73.6 ± 10.2	177 ± 9						
Douglas et al. (2018)	36	Males & females	24 track & field athletes	International athletes	24 ± 6	73 ± 10	178 ± 9	All physically active > 1 year	DJ @ 25 cm	FT/GCT (ms/ms)	Sprint (TF) vs. recreational (Re) athletes	TF = 3.08 ± 0.4 vs. Re = 2.11 ± 0.25	3.64 very large
			12 recreational	Recreationally active	26 ± 3	73 ± 11	173 ± 8		DJ @ 50 cm			TF = 3.08 ± 0.5 vs. Re = 2.07 ± 0.36	2.65 very large
									DJ @ 75 cm			TF = 2.9 ± 0.53 vs. Re = 1.88 ± 0.37	2.56 very large

Table 3 (*Continued*)

Study	Participant characteristics								RSI	Outcome measures			
	n	Sex	Sport	Competition level	Age (yrs)	Body mass (kg)	Height (cm)	Training status	Test	Calculation method	Population comparison	RSI value (units displayed in calculation column)	Effect size
Freitas et al. (2018)	24	Males	13 rugby union — backs	Elite	22 ± 2.3	84 ± 7.9	179.2 ± 7.3	ND	DJ @ 45 cm	JH/GCT (m/s)	Back line vs. forward line positions in rugby	Backs = 1.35 ± 0.52 vs. forwards = 0.99 ± 0.23	0.84 moderate
			11 rugby union — forwards		23.1 ± 3.2	108.3 ± 12.8 kg	185.6 ± 8.8						
											Faster vs. slower change of direction deficit	F = 1.24 ± 0.24 vs. S = 1.01 ± 0.29	0.83 moderate
Healy et al. (2019)	28	14 males	Sprint athletes	International and national athletes	22 ± 2	73.1 ± 6.8	182 ± 7	> 2 years S&C experience	DJ @ 30 cm	JH/GCT (m/s)	Male vs. female sprinters	M = 2.06 ± 0.43 vs. Fm = 1.65 ± 0.35	1.04 (0.26–1.82) moderate
		14 females			22 ± 4	64.4 ± 4.6	172 ± 7						
Jones et al. (2016)	27	Females	15 rugby league — backs	Elite	23.5 ± 4.1	66 ± 7.3	163.1 ± 4	No current structured S&C training	DJ @ 30 cm	JH/GCT (m/s)	Back line vs. forward line positions in rugby	Backs = 0.87 ± 0.31 vs. forwards = 0.58 ± 0.13	1.17 moderate
			12 rugby league — forwards		26.3 ± 6.4	80.7 ± 14.3	167.4 ± 6.8						
Keiner et al. (2015)	163	ND	Soccer	50 pro soccer players	26.1 ± 3.7	77.7 ± 6.5	183.8 ± 6.9	No current structured S&C training	DJ @ 24 cm	JH/GCT (cm/s)	Professional vs. semi-professional and academy level soccer players	Pro = 203 ± 32	Pro vs. Am/U17 = 0.7– 0.9 large
				20 semi-pro	22.1 ± 1.9	67.3 ± 3.5	177.9 ± 3.6					Semi = 187 ± 23	
				21 amateur	24.1 ± 3.7	74.1 ± 4.7	179.2 ± 3.6					Am = 157 ± 48	
				39 U19 players	18.4 ± 0.5	72.8 ± 7.2	177.9 ± 5.9					U19 = 189 ± 34	
				33 U17 players	16.7 ± 0.2	71.3 ± 7.6	179.3 ± 6.1					U17 = 172 ± 26	

Table 3 (*Continued*)

Study	Participant characteristics								RSI	Outcome measures			
	n	Sex	Sport	Competition level	Age (yrs)	Body mass (kg)	Height (cm)	Training status	Test	Calculation method	Population comparison	RSI value (units displayed in calculation column)	Effect size
Kobal et al. (2017)	79	ND	21 soccer players	National & international athletes	21.8 ±3	77.6 ±7	180.3 ±6.2	1–2 strength sessions/week	DJ @ 45 cm	JH/GCT (ND)	Variance between different sports	ND	Soccer vs. power = −3.81 very large
			24 tennis players		24.7 ±6	72.4 ±7.8	177.4 ±7						Tennis vs. power = −3.98 very large
			15 endurance runners		24.1 ±3.3	59.6 ±7.1	175.4 ±2.1						Endurance vs. power = −3.35 very large
			19 power athletes (sprinters & jumpers)		23.3 ±5	78.5 ±6.2	179.2 ±2.6	3–4 strength sessions/week					
Legg et al. (2017)	29	Females	9 basketball – guards	Australian women's international team	24.1 ±3.6	67.01 ±6.11	173.93 ±6.52	ND	DJ @ 30 cm	JH/GCT (m/s)	Positional differences in Women's basketball	1.31 ±0.32	ND
			9 basketball – forwards			77.03 ±4.16	184.95 ±5.39					1.41 ±0.52	
			11 basketball centres			78.66 ±7.16	190.09 ±4.19					1.24 ±0.30	
Lockie et al. (2011)	20	Males	Recreational sport	Recreationally active	21.9 ±2.9	80.5 ±8.5 kg	181 ±7	> 12 month strength training @ 2×/week	DJ @ 40 cm	JH/GCT (m/s)	Faster (4.93 m/s) vs. slower (4.53 m/s) participants in 10 m sprint test	Faster = 1.38 vs. slower = 0.93	1.61 very large

Table 3 (*Continued*)

Study	Participant characteristics								RSI	Outcome measures			
	n	Sex	Sport	Competition level	Age (yrs)	Body mass (kg)	Height (cm)	Training status	Test	Calculation method	Population comparison	RSI value (units displayed in calculation column)	Effect size
Loturco et al. (2015)	41	ND	22 power athletes	Olympic, Pan-American & National Brazilian Athletes	27.2 ±3.6	79.4 ±8.6	180.2 ±5.4	ND	DJ @ 45 cm		Power (sprinters, jumpers & throwers) vs. endurance athletes	2.19 ±0.58	ND
			19 endurance athletes		27.1 ±6.9	62.2 ±13.1	169.9 ±9.8					0.84 ±0.39	
Mooney et al. (2019)	54	ND	Gaelic football	40 U20 inter-country squad	19.1 ±0.6	78.1 ±6.6	181 ±5	ND	DJ @ 30 cm	JH/GCT (m/s)	Elite vs. sub-elite	1.9 ±0.38	0.98 large
				14 invited for inter-country trials	19.2 ±0.5	82.1 ±9.6	181 ±4					1.5 ±0.33	
Pehar et al. (2017)	110	Males	Basketball	58 division 1 athletes	21.58 ±3.92	89.33 ±10.91	194.92 ±8.09	ND	DJ @ 40 cm	JH/GCT (m/s)	1st vs. 2nd division teams	1.53 ±0.35	−0.28 small
				52 division 2 athletes								1.63 ±0.37	
			Guards			81.69 ±6.33	188.23 ±5.52				Positional difference in men basketball	1.69 ±0.32	ND
			Forwards			90.81 ±6.1	197.1 ±5.09					1.61 ±0.42	
			Centres			97.78 ±10.94	201.72 ±5.26					1.41 ±0.32	
Pojskic et al. (2018)	20	Males	Soccer	Highest 1st grade Swedish junior team	17 ±0.9	70.05 ±7.41	181 ±3	ND	DJ @ 25 cm	JH/GCT (m/s)	Variance between U17 & U19 soccer athletes	U17 = 1.47 ±0.23 vs. U19 = 1.52 ±0.16	$P \geq 0.05$ ES = 0.23 small
Sattler et al. (2015)	73	39 males	ND	College athletes	21.9 ±1.9	80.66 ±7.69	182.95 ±5.19	ND	DJ @ 40 cm	JH/GCT (ND)	High vs. average vs. low achievers based of CoD and RAT	CoD = HA = 39.19 ±3.67, AA = 33.43 ±4.02, LA = 35.79 ±3.89	ND
		34 females				61.95 ±6.7	171.45 ±6.81					RAT= HA = 35.79 ±4.76, AA = 35.21 ±5.23, LA = 35.12 ±3.12	ND

Table 3 (*Continued*)

Study	Participant characteristics								RSI	Outcome measures			
	n	Sex	Sport	Competition level	Age (yrs)	Body mass (kg)	Height (cm)	Training status	Test	Calculation method	Population comparison	RSI value (units displayed in calculation column)	Effect size
Sekulic et al. (2021)	75	Males	Futsal	Professional Futsal players – National level Croatia, Bosnia & Herzegovina	25.1 ±5.1	80.8 ±10.4	182.3 ±6.2	ND	DJ @ 40 cm	JH/GCT (cm/s)	Players who started the season vs. non-starters	St = 159.25 ±37.86 vs. NSt = 134.35 ±33.39	Moderate
Thomas et al. (2017)	16	Males	ND	ND	17.3 ±0.6	80.7 ±10	186.1 ±9.8	2×/week strength training @ time of study	DJ @ 30 cm	FT/GCT (ms/ms)	Stronger athletes (31.75 N/kg) vs. weaker athletes (25.45 N/kg)	Stronger = 1.42 ±0.28 vs. weaker = 1.28 ±0.23	$P \geq 0.005$ ES = 0.55 small
Wilkinson et al. (2012)	31	20 males	Squash	English performance program	22.6 ±1.3	72.9 ±5.2	ND	ND	DJ @ 30 cm	JH/GCT (cm/s)	Males vs. females		0.7 moderate
		11 female			22 ±1.3	62.3 ±4.6							
				Senior	ND	ND	ND				Senior vs. transition vs. TASS players	M = 291 ±45 vs. Fm = 250 ±31	S vs. TASS = 1.1 large; T vs. TASS = 1.0 large
				Transition								M = 294 ±51 vs. Fm = 252 ±56	
				Students on pro schedule								M = 235 ±54 vs. Fm = 186 ±21	

RSI: Reactive Strength Index; n: participant sample size; ND: not displayed; ES: effect size; DJ: drop jump; S: stronger; W: weaker; JH: jump height; GCT: ground contact time; FT: flight time; F: faster; S: slower; TS: team sport; TF: track & field; M: male; Fm: female; CoD: change of direction test; RAT: Reactive Agility Test; HA: high achievers, AA: average achievers; LA: low achievers; St: starters; NSt: non-starters.

values of 2.06 ± 0.43 and 291 ± 45, whilst women had comparative RSI scores of 1.65 ± 0.35 and 250 ± 31, respectively.

3.4.2. High vs. low achievers and elite vs. sub-elite

Thirteen studies ($n = 572$) compared high vs. low or elite vs. sub-elite populations in sport. High and low achievers (6 studies) could be participants of the same competition level (could be recreational or elite) or a pooled population of different competition levels. They were then divided into high and low achiever groups based on their performance on a specific outcome measure of the study, i.e. isometric mid-thigh pull (IMTP) strength results ($n = 2$), sprint time over 5 m to 40 m ($n = 3$) and change of direction and agility tests ($n = 1$). This was typically done in statistical analysis portion of paper. Elite vs. sub-elite populations (7 studies) were already separated populations based on competition level. Seven studies found large to very large differences between high and low achievers/ elite vs sub-elite, 2 studies found moderate differences, 1 study found small insignificant differences, whilst 3 studies found no difference at all.

3.4.3. Different sports

Three studies ($n = 144$) explored the difference in RSI between different sporting disciplines. Sprint and power athletes (jumpers and throwers) had significantly greater RSI scores compared to other sporting disciplines such as tennis, soccer, endurance running, hockey and rugby with moderate to very large effect sizes (0.73 to 3.98). Sprint and power athlete RSI scores ranged between 2.19 to 2.98 compared to 0.84 to 2.71 scores for the aforementioned sporting disciplines. All participants were national and international athletes with extensive training history.

3.4.4. Different playing positions

Four studies ($n = 187$) examined the difference of RSI between athletes who play different positions in particular sports. Two studies examined basketball and two examined rugby league. Two studies found moderate (ES = 0.84—1.17) differences in RSI between forwards and backs in rugby league. The other two studies did not report effect size, however one found insignificant differences in RSI between centres (1.24 ± 0.30), forwards (1.41 ± 0.52) and guards (1.31 ± 0.32), whilst the other found significant differences in RSI between centres (1.41 ± 0.32) and guards/forwards (1.61—1.69 ± 0.32—0.42). In all studies, positions that require athletes to have high body mass had an inferior RSI score than opposing positions.

4. Discussion

The present review examined whether RSI was a discriminating factor between populations. From the articles reviewed, it appears that RSI can differentiate between population groups with predominately moderate to large effect. It indicates that stretch-shortening cycle capacity is an important neuromuscular function, and that RSI can be used to differentiate populations' athleticism. This is important as locomotion is often a key component of athleticism, and whilst there may be other measurements that can be used to assess athletic performance, RSI is a practical measure of SSC capacity.

4.1. Males and females

Our findings suggest that males had moderately greater RSI than females. This difference has been attributed to the variation in strength capacity between males and females. Healy et al. [24] found that men had significantly greater relative peak force than females (36.3 ± 6.2 N vs. 29.8 ± 5.2 N), and subsequently found that men had moderately greater RSI than women (2.06 ± 0.43 vs. 1.65 ± 0.35). This difference in strength is due to the larger cross-sectional area of muscle fibres found in males compared to females, which has been seen as some 25—45% smaller in vastus lateralis and biceps brachii in females with varying levels of physical activity history [25]. Strength improves the capacity to able to tolerate high eccentric stretch loads, which then allows for quicker isometric muscle contraction of the muscle creating greater stiffness of the muscle-tendon unit of the lower limb (more specifically of the calf) [26]. These neuromuscular requirements are needed in high amplitude plyometric movements such as the drop jump. Therefore, males have moderately greater RSI than females as they have greater relative strength, an important factor to be able to quickly absorb upon landing before transferring to the concentric take off phase of the drop jump.

4.2. High vs. low achievers and elite vs. sub-elite

Overall, RSI appears to differentiate between high vs. low achievers of the same outcome measure, and elite vs. sub-elite athletes of the same sport. It was found that participants who excelled in their sport typically attained large RSI scores compared to their inferior counterpart. These differences in RSI scores may be due to several reasons. Large differences in RSI found in elite athletes compared to sub-elite athletes demonstrates greater skill set development in sport specific movement tasks, such as change of direction, sprinting, etc. [27,28]. This may attribute to RSI development; however, the cause and effect of this relationship couldn't be determined from these studies. Furthermore, elite athletes appear to be stronger than sub-elite athletes, represented by greater RSI scores at high drop heights, which demonstrates the importance of strength to remain stiff upon landing to minimise contact time at increasing higher eccentric demand [13]. Only one study highlighted that greater training volume may contribute to greater RSI scores, with U17 academy soccer players having larger RSI than amateur soccer players potentially due to their weekly training volume (7.5—12 hrs/wk vs. 3—6 hrs/week respectively) [3]. Although none of the studies reviewed highlighted the strength training schedule for elite and sub-elite athletes, it is plausible that elite athletes have a more consistent and structured training program, which would allow for greater development of physical characteristics such as strength and speed, which would improve reactive strength performance.

The two studies that used IMTP peak force as an outcome measure found that stronger athletes had a better RSI than their weaker counterparts, however, one study

found moderate to large differences [11], whilst the other found small insignificant differences [29]. There may be a few reasons that contribute to this. Firstly, the IMTP results from the Thomas et al. [29] (stronger = 31.75 N/kg vs. weaker = 25.45 N/kg) were lower than that of Beattie et al. [11] (stronger = 49.59 N/kg vs. weaker = 33.06 N/kg, respectively). The insignificant small difference in RSI found in Thomas et al. [29], may be due to the lower strength capacity of the participants, meaning that even the stronger participants did not have the adequate strength to handle the eccentric load in the drop jump. This is supported by the age (17.3 ± 0.6 years) of the participants in Thomas et al. [29] study, where strength training age is likely minimal compared to the average age in Beattie et al. [11] study (23.7 ± 4 years). Another reason is that Beattie et al. [11] split the participants into quartiles based off their IMTP results and analysed the top and bottom performing quartiles, whilst Thomas et al. [29] split the population in half. Therefore, the characteristics of the two groups in Thomas et al. [29] study will likely have greater similarities due to the overlapping standard deviation between the two groups.

Three studies split their participants in half based off fastest sprint times. Each study used different sprint distances of 5 m [30], 10 m [31] and 40 m [32]. Faster athletes in all three studies displayed higher RSI, however, there was a differentiation on the strength of difference. Moderate and large RSI differences were found when comparing 40 m and 10 m respectively, which may be due to force production efficiency during contact time. This was found in Lockie et al. [31] study where faster 10 m sprinters had reduced ground contact times (GCT) (250 ± 70 ms vs. 290 ± 80 ms) and greater jump heights (33 ± 6 cm vs. 27 ± 11 cm). The combination of smaller contact time and greater jump height is another example of participants having high eccentric tolerance, with faster athletes able to create a larger impulse on ground contact to transfer into the concentric toe off phase of jumping more efficiently [12]. Although participants who ran faster over 5 m had larger RSI than their slower counterpart (180 cm/s vs. 158 cm/s), the insignificance may be attributed to the distance that was tested, as a 5 m distance assesses first step quickness rather than acceleration which is more closely aligned with 10 m sprint times [31].Overall, the temporal similarities in ground contact time in the 10m sprint and RSI may be why this distance had the strongest relationships, as the 5m sprint is associated with higher force generation and longer GCT, whilst the 40m sprint is associated with shorter GCT.

One study examined high and low achievers based on both change of direction and reactive agility tests [33]. It was found that high achievers in well-trained college athletes had significantly greater RSI than their low achiever counterparts (39.19 ± 3.67 vs. 35.79 ± 3.89) when comparing results of change of direction tests, however there was no significant difference between high and low achievers when comparing reactive agility results (35.79 ± 4.76 vs. 35.12) [33]. This difference in result highlights the different physical and cognitive demands change of direction tasks and reactive agility tasks have, with pre-planned routes allowing for less cognitive delay due to decision making, allowing for quicker neuromuscular outputs. This pre-planned motor control is similar during drop jump tests.This concurs with previous research that found reactive strength had relationships with change of direction speed but not defensive agility performance [34].

Overall, strength appears to be a key underlying indicator for separating high and low achievers. This is then represented in RSI performance with stronger participants able to create a larger impulse during the contact phase of the drop jump, allowing for shorter contact time and greater jump height. However, specificity of reactive strength to the outcome measure (i.e. sprint speed) must be understood to be able to use RSI as a discriminating factor for performance.

4.3. Different sports

It is apparent in our findings that sprint and power athletes have superior RSI performance (moderate to large effect sizes) as opposed to other sports. These highly specific sports require explosive training, and as a result, athletes are frequently exposed to tasks that require high neuromuscular demands through specific sport training (sprinting, throwing and jumping) and strength training. This was seen in one study which highlighted that sprint and power athletes strength-trained 3—4 times per week whilst all other athletes (soccer, tennis and endurance) strength-trained 1—2 times per week [35]. All sports analysed have physical aspects that require reactive strength (change of direction, running economy, sprint acceleration) and therefore is apparent that sprint and power athletes may have partly developed the strength and stiffness requirements through larger resistance training volume. Thus, improving eccentric tolerance upon landing and force transfer into flight [12]. However, without insight into the strength programs, it can only be assumed from previous literature that the strength programs from all sports incorporated some level of max strength, power and plyometric development [36—40]. The fact that only assumptions of strength programs can be made is a consideration upon reviewing the Kobal et al. [35] study. Sprinters also have the additional benefit of having frequent exposure to max velocity sprinting where ground contact time is at its lowest and repeated fSSC actions occur, which mimic drop jump requirements.

A further consideration is how much influence genetics has on sport specialisation. It is possible that a higher type-1 muscle fibre composition in endurance athletes may limit force production capability [41], whilst a higher proportion (70%) of muscle fibre composition in sprinters is made up of type-2 fibres [42,43]. Therefore, it is also possible that greater RSI scores in power athletes may be equally attributed to naturally stronger and faster athletes, and therefore, impact of training may not be as significant.

4.4. Different playing positions

Of the four papers that analysed the influence of player position on reactive strength, it appears that positions that require athletes to have a high body weight have decreased reactive strength ability. Significant differences in weight between forward line and back line players in both elite male (108.3 ± 12.8 vs. 84 ± 7.9 kg) and female (80.7 ± 14.3 vs. 66 ± 7.3 kg) rugby also found significant moderate difference in RSI (0.58 ± 0.13—0.99 ± 0.23 vs. 0.87 ± 0.31—1.35 ± 0.52 respectively) [32,44]. Significant

differences in weight also reflected RSI performance in RSI between centres and guards/forwards in professional male (weight = 97.78 ± 10.94 vs. 81.69 ± 6.33/90.81 ± 6.1 kg; RSI = 1.41 ± 0.32 vs. 1.69 ± 0.32/1.61 ± 0.42) and female (weight = 78.66 ± 7.16 vs. 67.01 ± 6.11/77.03 ± 4.16 kg; RSI = 1.24 ± 0.30 vs. 1.31 ± 0.32/1.41 ± 0.52) basketball [45,46]. A reason why these heavier athletes had decreased reactive strength performance was due to an increased force requirement or time needed to resist the larger momentum [45]. This was a similar conclusion by Legg et al. [46] which stated that inadequate maximal relative strength in heavier athletes limited ability to decrease momentum and remain stiff during impact phase of ground contact, reducing ability to produce force for jump height. In elite female rugby league, the significantly greater total fat mass in forward line players negatively affected reactive strength performance [44]. This would impact relative strength due to a decreased lean muscle mass relative to total weight. This is potentially why moderate differences were found in rugby and not in basketball, as forward line players may have greater fat mass, as opposed to centres in basketball, who weigh more, but may be significantly leaner. This would be an area for future research. Furthermore, this may also support the moderate differences in RSI between males and females, as women in both studies were lighter than the men [24,28], as well as the previously supported research that females have higher fat mass [25,47], combine to minimise total lean muscle mass in females, reducing relative strength and therefore, reactive strength capacity.

Although not assessed, another aspect that may need to be considered is the physical demands that different sporting positions require that could affect the amount of exposure the athletes have to reactive movements. For instance, in rugby league forward line players typically run in a straight line, and cover less total and high-speed distance opposed to backline players, which may influence their lower RSI scores [48]. Overall, it appears that positions that require greater body mass have decreased reactive strength efficiency due to their reduced relative strength, which may be influenced by greater fat mass.

4.5. Limitations, practical recommendations and future research

Several factors should be considered when interpreting this review. Firstly, this review merely provides a qualitative conclusion of the findings from the available literature and does not provide a quantitative comparison through a meta-analysis. Hence, why effect sizes or means were only compared between applicable studies and not all studies. However, whilst findings may be relative to each study, trends of results can still be observed across all studies and summarised, which has been done in this review. Second, the cohort analysis used in these articles cannot determine a causal relationship, therefore, differences in RSI between populations cannot be correlated with their individual performances. This was a major limitation mentioned in the majority of studies analysed in this review. Third, the different units of measurements used to represent ground contact time and flight time (seconds or milliseconds) or jump height (centimetres or metres) varied between studies, meaning RSI values were inconsistent, and therefore clear standardised values could not be determined from this review. Fourth, it is worth considering the five studies that did not specify sex of their participants may have an impact on normative values for their RSI and other strength scores. However, these studies were used to determine difference between competition levels and between different sports, and when assuming equal participation of both males and females in these studies, the trends of the results will remain the same. Finally, 10 different box heights were used across the 19 studies to determine RSI from the DJ, further adding to the inability to standardise RSI results and evenly compare results across different studies. Regarding participant descriptions, only 5/19 studies mentioned level of strength training experience, an important piece of information regarding neuromuscular development. Even if studies explored max strength results, an indication of training history can help determine if athletes have achieved the top of their max strength capacity or whether they are still developing.

Whilst strength appears to be an important indicator of RSI ability, it is also known that plyometric training is an important training modality in improving RSI performance. Future research should compare whether strength training is more effective than plyometric training at developing RSI across different populations. This may be beneficial as plyometric exercise may be an important progression for populations with adequate relative strength but may have diminishing returns in weaker populations with their inability to control bodyweight. Additionally, whilst variance in playing position was reviewed in the paper and was heavily influenced by anthropometrical differences such as height and weight, future research should investigate comparing populations based on their height and weight to determine if these anthropometrical characteristics differentiate RSI performance.

5. Conclusion

The aim of this review was to assess whether there was any significant difference in RSI between different populations. It was determined that there was a range of differences between different populations. Sprint and power athletes had moderate to large increase in RSI than team sport athletes, males had moderately higher RSI than females, high achievers typically had large RSI opposed to low achievers and positions that required athletes to be heavier tended to have small to moderately lower RSI than lighter athletes. A contributing factor for these differences in RSI scores is due to the relative strength of the participants. The higher the relative strength of the population, the greater the RSI score. The relative strength of the athlete represents their ability to be eccentrically strong enough to control their bodyweight on ground contact, which allows for more efficient force production and therefore greater jump height. Strength and conditioning practitioners should incorporate strength work aimed to develop athlete's relative strength to assist in reactive strength development.

Funding

No funding was received for this work.

Disclosure of interest

The authors declare that they have no competing interest.

Data availability

The data within this systematic review is secondary data available through the relevant articles referenced throughout.

Acknowledgments

Authors would like to thank Kim Southey and Peter Harrison for their involvement in this study.

References

[1] Bosco C, Ito A, Komi PV, Luhtanen P, Rahkila P, Rusko H, et al. Neuromuscular function and mechanical efficiency of human leg extensor muscles during jumping exercises. Acta Physiol Scand 1982;114(4):543—50.
[2] Hennessy L, Kilty J. Relationship of the stretch-shortening cycle to sprint performance in trained female athletes. J Strength Cond Res 2001;15(3):326—31.
[3] Keiner M, Sander A, Wirth K, Hartmann H. Differences in the performance tests of the fast and slow stretch and shortening cycle among professional, amateur and elite youth soccer players. J Hum Sport Exercise 2015;10(2):563—70.
[4] Komi PV, Bosco C. Utilization of stored elastic energy in leg extensor muscles by men and women. Med Sci Sports 1978;10(4):261—5.
[5] Turner AN, Jeffreys I. The stretch-shortening cycle: proposed mechanisms and methods for enhancement. Strength Cond J 2010;32(4):87—99.
[6] Farley CT, Blickhan R, Saito J, Taylor CR. Hopping frequency in humans: a test of how springs set stride frequency in bouncing gaits. J Appl Physiol 1991;71(6):2127—32.
[7] Van Hooren B, Zolotarjova J. The difference between countermovement and squat jump performances: a review of underlying mechanisms with practical applications. J Strength Cond Res 2017;31(7):2011—20.
[8] Nummela A, Keränen T, Mikkelsson LO. Factors related to top running speed and economy. Int J Sports Med 2007;28(8):655—61.
[9] Mattes K, Wolff S, Alizadeh S. Kinematic stride characteristics of maximal sprint running of elite sprinters — Verification of the ''Swing-Pull Technique''. J Hum Kinet 2021;77(1):15—24.
[10] García-Pinillos F, Lago-Fuentes C, Latorre-Román PA, Pantoja-Vallejo A, Ramirez-Campillo R. Jump-rope training: improved 3-km time-trial performance in endurance runners via enhanced lower-limb reactivity and foot-arch stiffness. Int J Sports Physiol Perform 2020;15(7):927—33.
[11] Beattie K, Carson BP, Lyons M, Kenny IC. The relationship between maximal strength and reactive strength. Int J Sports Physiol Perform 2017;12(4):548—53.
[12] Douglas J, Pearson S, Ross A, McGuigan M. Reactive and eccentric strength contribute to stiffness regulation during maximum velocity sprinting in team sport athletes and highly trained sprinters. J Sports Sci 2020;38(1):29—37.
[13] Douglas J, Pearson S, Ross A, McGuigan M. Kinetic determinants of reactive strength in highly trained sprint athletes. J Strength Cond Res 2018;32(6):1562—70.
[14] Fei L, Newton RU, Yue S, Sutton D, Haiyong D. Correlation of eccentric strength, reactive strength, and leg stiffness with running economy in well-trained distance runners. J Strength Cond Res 2021;35(6):1491—9.
[15] Schmidtbleicher D. Training for power events. In: Komi PV, editor. Strength & power in sport. Oxford, United Kingdom: Blackwell Science; 1992. p. 169—79.
[16] Flanagan EP, Comyns TM. The use of contact time and the reactive strength index to optimize fast stretch-shortening cycle training. Strength Cond J 2008;30(5):32—8.
[17] Barker LA, Harry JR, Mercer JA. Relationships between countermovement jump ground reaction forces and jump height, reactive strength index, and jump time. J Strength Cond Res 2018;32(1):248—54.
[18] Cormie P, McGuigan MR, Newton RU. Changes in the eccentric phase contribute to improved stretch-shorten cycle performance after training. Med Sci Sports Exerc 2010;42(9):1731—44.
[19] Kuitunen S, Komi PV, Kyröläinen H. Knee and ankle joint stiffness in sprint running. Med Sci Sports Exerc 2002;34(1):166—73.
[20] Chumanov ES, Heiderscheit BC, Thelen DG. Hamstring musculotendon dynamics during stance and swing phases of high-speed running. Med Sci Sports Exerc 2011;43(3):525—32.
[21] Jarvis P, Turner A, Read P, Bishop C. Reactive strength index and its associations with measures of physical and sports performance: a systematic review with meta-analysis. Sports Med 2022;52(2):301—30.
[22] Moher D, Liberati A, Tetzlaff J, Altman DG. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. BMJ 2009;339(7716):e78—336.
[23] Downs SH, Black N. The feasibility of creating a checklist for the assessment of the methodological quality both of randomised and non-randomised studies of health care interventions. J Epidemiol Community Health 1998;52(6):377—84.
[24] Healy R, Smyth C, Kenny IC, Harrison AJ. Influence of reactive and maximum strength indicators on sprint performance. J Strength Cond Res 2019;33(11):3039—48.
[25] Miller AEJ, MacDougall JD, Tarnopolsky MA, Sale DG. Gender differences in strength and muscle fiber characteristics. Eur J Appl Physiol Occup Physiol 1993;66(3):254—62.
[26] Lichtwark G, Bougoulias K, Wilson A. Muscle fascicle and series elastic element length changes along the length of the human gastrocnemius during walking and running. J Biomech 2005;40(1):157—64.
[27] Mooney TJ, Malone S, Cullen BD, Darragh I, Bennett S, Collins K, et al. Investigating the role of anthropometric and physical performance measures on team selection in elite and sub-elite under-20 Gaelic football players. J Austr Strength Cond 2019;27(5):14—24.
[28] Wilkinson M, Cooke M, Murray S, Thompson KG, Clair Gibson AST, Winter EM. Physiological correlates of multiple-sprint ability and performance in international-standard squash players. J Strength Cond Res 2012;26(2):540—7.
[29] Thomas C, Kyriakidou I, Dos'Santos T, Jones PA. Differences in vertical jump force-time characteristics between stronger and weaker adolescent basketball players. Sports 2017;5(3):63—73.
[30] Cronin JB, Hansen KT. Strength and power predictors of sports speed. J Strength Cond Res 2005;19(2):349—57.
[31] Lockie RG, Murphy AJ, Knight TJ, De Jonge XAKJ. Factors that differentiate acceleration ability in field sport athletes. J Strength Cond Res 2011;25(10):2704—14.
[32] Freitas TT, Alcaraz PE, Bishop C, Calleja-Gonzalez J, Arruda AFS, Guerriero A, et al. Change of direction deficit in national team rugby union players: is there an influence of playing position? Sports 2019;7(1).

[33] Sattler T, Sekulic D, Spasic M, Peric M, Krolo A, Uljevic O, et al. Analysis of the association between motor and anthropometric variables with change of direction speed and reactive agility performance. J Hum Kinet 2015;47(1):137—45.
[34] Young WB, Miller IR, Talpey SW. Physical qualities predict change-of-direction speed but not defensive agility in Australian rules football. J Strength Cond Res 2015;29(1):206—12.
[35] Kobal R, Nakamura FY, Kitamura K, Cal Abad CC, Pereira LA, Loturco I. Vertical and depth jumping performance in elite athletes from different sports specialties. Sci Sports 2017;32(5):e191—6.
[36] Barber-Westin SD, Hermeto AA, Noyes FR. A six-week neuromuscular training program for competitive junior tennis players. J Strength Cond Res 2010;24(9):2372—82.
[37] Blazevich AJ, Jenkins DG. Effect of the movement speed of resistance training exercises on sprint and strength performance in concurrently training elite junior sprinters. J Sports Sci 2002;20(12):981—90.
[38] Ferley DD, Scholten S, Vukovich MD. Combined sprint interval, plyometric, and strength training in adolescent soccer players: effects on measures of speed, strength, power, change of direction, and anaerobic capacity. J Strength Cond Res 2020;34(4):957—68.
[39] Kraemer WJ, Häkkinen K, Gordon SE, Cummings D, Hauth J, Pullo F, et al. Physiological changes with periodized resistance training in women tennis players. Med Sci Sports Exerc 2003;35(1):157—68.
[40] Taipale RS, Mikkola J, Vesterinen V, Nummela A, Häkkinen K. Neuromuscular adaptations during combined strength and endurance training in endurance runners: maximal versus explosive strength training or a mix of both. Eur J Appl Physiol 2012;113(2):325—35.
[41] Harber MP, Gallagher PM, Trautmann J, Trappe SW. Myosin heavy chain composition of single muscle fibers in male distance runners. Int J Sports Med 2002;23(7):484—8.
[42] Trappe S, Luden N, Minchev K, Raue U, Jemiolo B, Trappe TA. Skeletal muscle signature of a champion sprint runner. J Appl Physiol 2015;118(12):1460—6.
[43] Powers SK, Howley ET. Exercise physiology: theory and application to fitness and performance. 9th ed. New York, NY: McGraw-Hill Education; 2015.
[44] Jones B, Emmonds S, Hind K, Nicholson G, Rutherford Z, Till K. Physical Qualities of international female rugby league players by playing position. J Strength Cond Res 2016;30(5):1333—40.
[45] Pehar M, Sekulic D, Sisic N, Spasic M, Uljevic O, Krolo A, et al. Evaluation of different jumping tests in defining position-specific & performance-level differences in high level basketball players. Biol Sport 2017;34(3):263—72.
[46] Legg JS, Williams KJ, Pyne DB, Semple SJ, Ball N. Physiological profile of Australian elite female basketball players according to playing position. J Austr Strength Cond 2017;25(2):6—12.
[47] Blaak E. Gender differences in fat metabolism. Curr Opin Clin Nutr Metab Care 2001;4(6):499—502.
[48] Glassbrook DJ, Doyle TLA, Alderson JA, Fuller JT. The demands of professional rugby league match-play: a meta-analysis. Sports Med Open 2019;5(1):24.